CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Faculty of Technology & Engineering

Devang Patel Institute of Advance Technology and Research (DEPSTAR)

CSUE301 – Big Data Analytics

Student ID: 24DIT021

Student Name: Hit Goyani

PRACTICAL 6

Practical 6

1. Problem Definition

Scenario:

ShopSmart Analytics Pvt. Ltd. wants to analyze its e-commerce sales data. The company needs to identify high-performing product categories, top-selling products and revenue contribution by city. Since the amount of data is large, Apache Spark is used for scalable data processing and analysis.

Objective

To use PySpark DataFrames to:

- Load e-commerce sales data
- Validate the dataset
- Calculate revenue
- Find category-wise revenue
- Find top-selling products
- Calculate city-wise revenue
- Sort categories according to revenue
- Generate analytical summaries
- Extract business insights

PART A — Practical Execution

Step 1: Creating Folder and files

Step 1: Creating Folder and Files

Welcome | sales_data.csv | practical6.py 2

Practical-6 > sales_data.csv > data

```
OrderID,Product,Category,Quantity,Price,City
O001,Laptop,Electronics,2,55000,Ahmedabad
O002,Mobile,Electronics,5,25000,Surat
O003,Headphones,Electronics,10,2000,Vadodara
O004,TShirt,Clothing,8,800,Ahmedabad
O005,Jeans,Clothing,5,1500,Surat
O006,Sofa,Furniture,2,30000,Ahmedabad
O007,Chair,Furniture,6,5000,Rajkot
O008,Watch,Accessories,7,2500,Surat
O009,Shoes,Footwear,4,3000,Vadodara
O010,Laptop,Electronics,1,55000,Rajkot
O011,TShirt,Clothing,10,800,Surat
O012,Sofa,Furniture,1,30000,Vadodara
O013,Mobile,Electronics,3,25000,Ahmedabad
O014,Shoes,Footwear,6,3000,Surat
O015,Watch,Accessories,4,2500,Rajkot
```

Problems 2 | Output | Debug Console | Terminal | Ports

```
PS C:\Users\ADMIN\OneDrive\Desktop\BDA> cd Practical-6
PS C:\Users\ADMIN\OneDrive\Desktop\BDA\Practical-6> python --version
Python 3.14.0
```

Practical-6 > practical6.py > ...

```python
from pyspark.sql import SparkSession
from pyspark.sql.functions import col, sum, avg

spark = SparkSession.builder \
    .appName("E-Commerce Sales Analysis") \
    .getOrCreate()

df = spark.read.csv("sales_data.csv", header=True, inferSchema=True)

print("Sales Data:")
df.show()
```

2. Output of Practical6.py

```
===== ORIGINAL DATA =====
+-------+----------+-----------+--------+-----+---------+
|OrderID|   Product|   Category|Quantity|Price|     City|
+-------+----------+-----------+--------+-----+---------+
|   0001|    Laptop|Electronics|       2|55000|Ahmedabad|
|   0002|    Mobile|Electronics|       5|25000|    Surat|
|   0003|Headphones|Electronics|      10| 2000| Vadodara|
|   0004|    TShirt|   Clothing|       8|  800|Ahmedabad|
|   0005|     Jeans|   Clothing|       5| 1500|    Surat|
|   0006|      Sofa|  Furniture|       2|30000|Ahmedabad|
|   0007|     Chair|  Furniture|       6| 5000|   Rajkot|
|   0008|     Watch|Accessories|       7| 2500|    Surat|
|   0009|     Shoes|   Footwear|       4| 3000| Vadodara|
|   0010|    Laptop|Electronics|       1|55000|   Rajkot|
|   0011|    TShirt|   Clothing|      10|  800|    Surat|
|   0012|      Sofa|  Furniture|       1|30000| Vadodara|
|   0013|    Mobile|Electronics|       3|25000|Ahmedabad|
|   0014|     Shoes|   Footwear|       6| 3000|    Surat|
|   0015|     Watch|Accessories|       4| 2500|   Rajkot|
+-------+----------+-----------+--------+-----+---------+
```

```
===== SCHEMA =====
root
 |-- OrderID: string (nullable = true)
 |-- Product: string (nullable = true)
 |-- Category: string (nullable = true)
 |-- Quantity: integer (nullable = true)
 |-- Price: integer (nullable = true)
 |-- City: string (nullable = true)

===== DATA TYPES =====
Column: OrderID      | Data Type: string
Column: Product      | Data Type: string
Column: Category     | Data Type: string
Column: Quantity     | Data Type: int
Column: Price        | Data Type: int
Column: City         | Data Type: string
```

Revenue Calculation and Product Analysis

```
===== REVENUE CALCULATION =====
+-------+----------+-----------+--------+-----+---------+-------+
|OrderID|   Product|   Category|Quantity|Price|     City|Revenue|
+-------+----------+-----------+--------+-----+---------+-------+
|   0001|    Laptop|Electronics|       2|55000|Ahmedabad| 110000|
|   0002|    Mobile|Electronics|       5|25000|    Surat| 125000|
|   0003|Headphones|Electronics|      10| 2000| Vadodara|  20000|
|   0004|    TShirt|   Clothing|       8|  800|Ahmedabad|   6400|
|   0005|     Jeans|   Clothing|       5| 1500|    Surat|   7500|
|   0006|      Sofa|  Furniture|       2|30000|Ahmedabad|  60000|
|   0007|     Chair|  Furniture|       6| 5000|   Rajkot|  30000|
|   0008|     Watch|Accessories|       7| 2500|    Surat|  17500|
|   0009|     Shoes|   Footwear|       4| 3000| Vadodara|  12000|
|   0010|    Laptop|Electronics|       1|55000|   Rajkot|  55000|
|   0011|    TShirt|   Clothing|      10|  800|    Surat|   8000|
|   0012|      Sofa|  Furniture|       1|30000| Vadodara|  30000|
|   0013|    Mobile|Electronics|       3|25000|Ahmedabad|  75000|
|   0014|     Shoes|   Footwear|       6| 3000|    Surat|  18000|
|   0015|     Watch|Accessories|       4| 2500|   Rajkot|  10000|
+-------+----------+-----------+--------+-----+---------+-------+
```

```
===== CATEGORY-WISE REVENUE =====
+-----------+-------------+
|   Category|Total_Revenue|
+-----------+-------------+
|Electronics|       385000|
|  Furniture|       120000|
|   Footwear|        30000|
|Accessories|        27500|
|   Clothing|        21900|
+-----------+-------------+

===== TOP-SELLING PRODUCTS =====
+----------+------------------+
|   Product|Total_Quantity_Sold|
+----------+------------------+
|    TShirt|                18|
|     Watch|                11|
|     Shoes|                10|
|Headphones|                10|
|    Mobile|                 8|
|     Chair|                 6|
|     Jeans|                 5|
|    Laptop|                 3|
|      Sofa|                 3|
+----------+------------------+
```

Discount Analysis and Least-Performing Category

```
===== DISCOUNT ANALYSIS =====
+-------+----------+-----------+--------+-----+---------+-------+--------+-------------+
|OrderID|   Product|   Category|Quantity|Price|     City|Revenue|Discount|Final_Revenue|
+-------+----------+-----------+--------+-----+---------+-------+--------+-------------+
|   O001|    Laptop|Electronics|       2|55000|Ahmedabad| 110000| 11000.0|      99000.0|
|   O002|    Mobile|Electronics|       5|25000|    Surat| 125000| 12500.0|     112500.0|
|   O003|Headphones|Electronics|      10| 2000| Vadodara|  20000|  2000.0|      18000.0|
|   O004|    TShirt|   Clothing|       8|  800|Ahmedabad|   6400|   640.0|       5760.0|
|   O005|     Jeans|   Clothing|       5| 1500|    Surat|   7500|   750.0|       6750.0|
|   O006|      Sofa|  Furniture|       2|30000|Ahmedabad|  60000|  6000.0|      54000.0|
|   O007|     Chair|  Furniture|       6| 5000|   Rajkot|  30000|  3000.0|      27000.0|
|   O008|     Watch|Accessories|       7| 2500|    Surat|  17500|  1750.0|      15750.0|
|   O009|     Shoes|   Footwear|       4| 3000| Vadodara|  12000|  1200.0|      10800.0|
|   O010|    Laptop|Electronics|       1|55000|   Rajkot|  55000|  5500.0|      49500.0|
|   O011|    TShirt|   Clothing|      10|  800|    Surat|   8000|   800.0|       7200.0|
|   O012|      Sofa|  Furniture|       1|30000| Vadodara|  30000|  3000.0|      27000.0|
|   O013|    Mobile|Electronics|       3|25000|Ahmedabad|  75000|  7500.0|      67500.0|
|   O014|     Shoes|   Footwear|       6| 3000|    Surat|  18000|  1800.0|      16200.0|
|   O015|     Watch|Accessories|       4| 2500|   Rajkot|  10000|  1000.0|       9000.0|
+-------+----------+-----------+--------+-----+---------+-------+--------+-------------+

===== LEAST-PERFORMING CATEGORY =====
+--------+------------------+
|Category|Total_Final_Revenue|
+--------+------------------+
|Clothing|           19710.0|
+--------+------------------+
only showing top 1 row
===== MONTHLY REVENUE =====
Monthly analysis requires an OrderDate column in the dataset. The dataset does not contain any date column.
PS C:\Users\ADMIN\OneDrive\Desktop\BDA\Practical-6> ERROR: The process with PID 18096 (child process of PID 21872) coul
d not be terminated.
Reason: There is no running instance of the task.
SUCCESS: The process with PID 21872 (child process of PID 2852) has been terminated.
SUCCESS: The process with PID 2852 (child process of PID 25932) has been terminated.
```

3. Create Revenue Column

```
===== REVENUE CALCULATION =====
+-------+----------+-----------+--------+-----+---------+-------+
|OrderID|   Product|   Category|Quantity|Price|     City|Revenue|
+-------+----------+-----------+--------+-----+---------+-------+
|   O001|    Laptop|Electronics|       2|55000|Ahmedabad| 110000|
|   O002|    Mobile|Electronics|       5|25000|    Surat| 125000|
|   O003|Headphones|Electronics|      10| 2000| Vadodara|  20000|
|   O004|    TShirt|   Clothing|       8|  800|Ahmedabad|   6400|
|   O005|     Jeans|   Clothing|       5| 1500|    Surat|   7500|
|   O006|      Sofa|  Furniture|       2|30000|Ahmedabad|  60000|
|   O007|     Chair|  Furniture|       6| 5000|   Rajkot|  30000|
|   O008|     Watch|Accessories|       7| 2500|    Surat|  17500|
|   O009|     Shoes|   Footwear|       4| 3000| Vadodara|  12000|
|   O010|    Laptop|Electronics|       1|55000|   Rajkot|  55000|
|   O011|    TShirt|   Clothing|      10|  800|    Surat|   8000|
|   O012|      Sofa|  Furniture|       1|30000| Vadodara|  30000|
|   O013|    Mobile|Electronics|       3|25000|Ahmedabad|  75000|
|   O014|     Shoes|   Footwear|       6| 3000|    Surat|  18000|
|   O015|     Watch|Accessories|       4| 2500|   Rajkot|  10000|
+-------+----------+-----------+--------+-----+---------+-------+
```

4. Calculate Category-wise Revenue

```
===== CATEGORY-WISE REVENUE =====
+-----------+-------------+
|   Category|Total_Revenue|
+-----------+-------------+
|Electronics|       385000|
|  Furniture|       120000|
|   Footwear|        30000|
|Accessories|        27500|
|   Clothing|        21900|
+-----------+-------------+
```

5. Find Top-Selling Products

```
===== TOP-SELLING PRODUCTS =====
+----------+------------------+
|   Product|Total_Quantity_Sold|
+----------+------------------+
|    TShirt|                18|
|     Watch|                11|
|     Shoes|                10|
|Headphones|                10|
|    Mobile|                 8|
|     Chair|                 6|
|     Jeans|                 5|
|    Laptop|                 3|
|      Sofa|                 3|
+----------+------------------+
```

6. Calculate City-wise Revenue

```
===== CITY-WISE REVENUE =====
+---------+-------------+
|     City|Total_Revenue|
+---------+-------------+
|Ahmedabad|       251400|
|    Surat|       176000|
|   Rajkot|        95000|
| Vadodara|        62000|
+---------+-------------+
```

7. Top 3 Revenue-Generating Cities and Average Revenue per Order

```
===== TOP 3 CITIES =====
+---------+-------------+
|     City|Total_Revenue|
+---------+-------------+
|Ahmedabad|       251400|
|    Surat|       176000|
|   Rajkot|        95000|
+---------+-------------+
only showing top 3 rows
===== AVERAGE REVENUE PER ORDER =====
+------------------------+
|Average_Revenue_Per_Order|
+------------------------+
|                 38960.0|
+------------------------+
```

8. Add Discount Column

```
===== DISCOUNT ANALYSIS =====
+-------+----------+-----------+--------+-----+---------+-------+--------+-------------+
|OrderID|   Product|   Category|Quantity|Price|     City|Revenue|Discount|Final_Revenue|
+-------+----------+-----------+--------+-----+---------+-------+--------+-------------+
|   0001|    Laptop|Electronics|       2|55000|Ahmedabad| 110000| 11000.0|      99000.0|
|   0002|    Mobile|Electronics|       5|25000|    Surat| 125000| 12500.0|     112500.0|
|   0003|Headphones|Electronics|      10| 2000| Vadodara|  20000|  2000.0|      18000.0|
|   0004|    TShirt|   Clothing|       8|  800|Ahmedabad|   6400|   640.0|       5760.0|
|   0005|     Jeans|   Clothing|       5| 1500|    Surat|   7500|   750.0|       6750.0|
|   0006|      Sofa|  Furniture|       2|30000|Ahmedabad|  60000|  6000.0|      54000.0|
|   0007|     Chair|  Furniture|       6| 5000|   Rajkot|  30000|  3000.0|      27000.0|
|   0008|     Watch|Accessories|       7| 2500|    Surat|  17500|  1750.0|      15750.0|
|   0009|     Shoes|   Footwear|       4| 3000| Vadodara|  12000|  1200.0|      10800.0|
|   0010|    Laptop|Electronics|       1|55000|   Rajkot|  55000|  5500.0|      49500.0|
|   0011|    TShirt|   Clothing|      10|  800|    Surat|   8000|   800.0|       7200.0|
|   0012|      Sofa|  Furniture|       1|30000| Vadodara|  30000|  3000.0|      27000.0|
|   0013|    Mobile|Electronics|       3|25000|Ahmedabad|  75000|  7500.0|      67500.0|
|   0014|     Shoes|   Footwear|       6| 3000|    Surat|  18000|  1800.0|      16200.0|
|   0015|     Watch|Accessories|       4| 2500|   Rajkot|  10000|  1000.0|       9000.0|
+-------+----------+-----------+--------+-----+---------+-------+--------+-------------+
```

9. Least-Performing Category

```
===== LEAST-PERFORMING CATEGORY =====
+--------+-------------------+
|Category|Total_Final_Revenue|
+--------+-------------------+
|Clothing|            19710.0|
+--------+-------------------+
only showing top 1 row
```

10. Monthly Revenue Summary

Our current dataset does not contain a date/month column.

So for this supplementary question, we would need to extend the dataset with an OrderDate column.

11. Key Questions / Analysis

Q1. Why are Spark DataFrames preferred over traditional spreadsheets for large-scale analytics?

Spark DataFrames are preferred because they can process very large datasets efficiently using distributed computing. Data can be divided into partitions and processed in parallel across multiple machines. Spreadsheets are more suitable for small or moderate-sized datasets and have limitations when the data becomes very large.

Q2. What is the role of the groupBy() operation in data aggregation?

The groupBy() operation groups records having the same value in a particular column. It is commonly used with aggregation functions such as sum(), avg(), count(), min() and max().

For example:
df.groupBy("Category").sum("Revenue")
groups the sales according to product category and calculates the total revenue for each category.

Q3. Explain the difference between transformations and actions in Spark.

Transformations create a new DataFrame or RDD from existing data. They are lazily evaluated.

Examples:
select()
filter()
withColumn()
groupBy()

Actions trigger the actual execution of Spark operations and return a result or output.

Examples:
show()
count()
collect()

Therefore:
Transformation → Defines the operation
Action → Executes the operation

Q4. How does Spark improve scalability for e-commerce analytics?

Spark improves scalability by using distributed and parallel processing. Large e-commerce datasets can be divided into partitions and processed simultaneously across multiple computing resources. Spark DataFrames also provide optimized execution, making large-scale sales analysis faster and more efficient.

Q5. Which category contributes the highest revenue and what business action can be taken?

The highest-revenue category is identified from the category-wise revenue analysis by sorting categories in descending order of revenue.

Highest Revenue Category: _____

A suitable business action is to maintain sufficient inventory for high-performing products, continue targeted marketing campaigns, and analyze customer demand to maintain the category's performance.

12. Conclusion

The practical successfully demonstrated the use of Apache Spark DataFrames for e-commerce sales analysis. The dataset was loaded and validated, and important business metrics such as revenue, category-wise sales, product quantity, and city-wise revenue were calculated.

The use of operations such as withColumn(), groupBy(), sum(), avg(), orderBy() and limit() helped perform efficient data analysis. Spark provides a scalable approach for processing large e-commerce datasets and generating useful business insights.

The analysis can help businesses identify high-performing categories and products, understand geographical revenue distribution, improve marketing strategies, and make data-driven decisions.